

Rational Design of Nonbonded Point Charge Models for Highly Charged Metal Cations with Lennard-Jones 12-6 Potential

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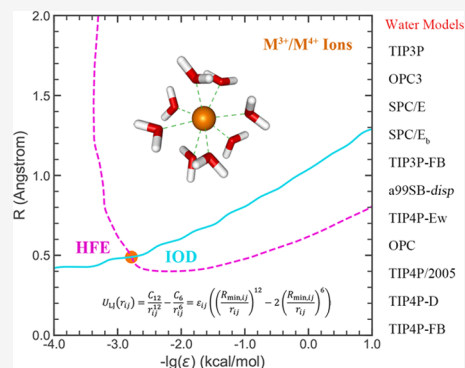
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ABSTRACT: Here, we developed nonbonded point charge models using a simple Lennard-Jones (LJ) 12-6 potential for highly charged metal cations (18 trivalent and 6 tetravalent ions) for use with 11 water models of TIP3P, OPC3, SPC/E, SPC/E_b, TIP3P-FB, a99SB-disp, TIP4P-Ew, OPC, TIP4P/2005, TIP4P-D, and TIP4P-FB. The designed models simultaneously reproduce the hydration free energy (HFE) and ion–oxygen distance (IOD) in the first hydration shell with an error within 1 kcal/mol and 0.01 Å on average, respectively, and yield reasonable coordination numbers for most cations. Such performance is equivalent to the previously reported point charge models using a more complex 12-6-4 LJ-type potential, while the LJ R parameters of our models are much close to Shannon's revised effective ion radii than that of the 12-6-4 models. Our designed models overestimate the diffusion constants of several trivalent ions by 5–68%. The performance in predicting osmotic coefficients of trivalent chlorides in aqueous solution depends on the salt type. A calibration of cation–anion interacting LJ parameters reproduces the experimental osmotic coefficients of an AlCl₃ solution at 0.2–3.0 mol/L. The effectiveness of our new models is further demonstrated by simulating a metalloprotein system with four force field/water combinations. This work facilitates accurate modeling of metal-containing systems by a variety of force fields and water models in aqueous solution.



INTRODUCTION

There is no doubt that highly charged (trivalent and tetravalent) metal ions play significant roles in biology^{1–4} and materials science.^{5–9} Acting as a cofactor, for instance, iron (Fe) participates in oxygen transport, oxygen storage, and energy generation¹⁰ and facilitates enzymatic reactions in DNA synthesis and repair.¹¹ Aluminum (Al) can affect the health of human beings, and aluminum exposure was reported to relate to neurodegenerative disorders such as Alzheimer's and Parkinson's diseases.¹² The lanthanide series can be used as efficient catalysts for organic synthesis like CO₂ conversion^{13–15} and as magnetic⁷ and optical materials.^{16–18} The actinide series always relate to radioactive contamination of environmental concern.¹⁹ These metals commonly exist in the form of highly charged ions, and an in-depth investigation, via experimental and/or computational techniques, is of crucial importance to explore the underlying molecular mechanism and reduce their harmfulness to the environment.

As a bridge between theory and experiment, computational simulation has been widely utilized to obtain valuable insights into a variety of complex mechanisms in metal-containing systems at the microscopic level.^{20–23} The simulation requires reliable force field (FF) models, and there are mainly three strategies for parameterizing the metal ions, including the bonded model,²⁴ the nonbonded (NB) point charge model,^{25–31} and the nonbonded dummy atom model.^{32–38} In the bonded model, predefined covalent bonds and/or

distance restrictions were constructed between the metal and its ligand, thereby reproducing the geometry of the metal center with high accuracy. This model does not allow ligand exchange, however, and such deficiency can be fixed by the nonbonded model. In the nonbonded dummy atom model, the dummy atoms bind covalently to the metal core in a specific coordination geometry, while they have no covalent bonds with the ligand. The number of dummy atom sites can be six^{32–36} or eight,³⁷ generally being the same as the metal coordination number (CN) of interest, although six dummy sites were able to produce a CN of 8 for Ca²⁺.³⁶

The nonbonded point charge model has attracted great attention in ion modeling because of its simplicity and computational efficiency. In this model, the ion is described as a point charge with a corresponding mass and has only Coulomb and van der Waals (vdW) interactions with the surroundings. If the charge of metal ions were fixed, only the parameters of vdW potential would need to be optimized, as did in a number of point charge models. Merz and co-workers

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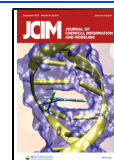


Table 1. Experimental Conventional ($\Delta G^{\text{o,conv}}$) and Real ($\Delta G^{\text{*,real}}$) Hydration Free Energy and Ion–Oxygen Distance (IOD) and Coordination Number (CN) in the First Hydration Shell for M(III) and M(IV) Metal Ions

metal	$\Delta G^{\text{o,conv}}$ (kcal/mol) ^a	$\Delta G^{\text{*,real}}$ (kcal/mol) ^b	IOD (Å) ^c	CN ^c	r_{ion} (Å) ^d	effective ion radii (Å) ^e
Al ³⁺	−326.2	−1107.5	1.887 ± 0.015	6	0.50	0.47
Cr ³⁺	−203.1	−984.4	1.969 ± 0.032	6	0.58	0.62
Fe ³⁺	−264.1	−1045.4	2.031 ± 0.019	6	0.64	0.61
In ³⁺	−196.0	−977.3	2.156 ± 0.002	6	0.76	0.78
Lu ³⁺	−84.8	−866.1	2.34	8	0.95	0.96
Tl ³⁺	−193.6	−974.9	2.231 ± 0.005	4–6	0.84	0.87
Tm ³⁺	−84.8	−866.1	2.36	8.1	0.97	0.98
Er ³⁺	−80.1	−861.4	2.363 ± 0.005	6.3–8.2	0.97	0.98
Y ³⁺	−69.3	−850.6	2.365	8	0.97	0.99
Dy ³⁺	−63.3	−844.6	2.370 ± 0.003	7.4–7.9	0.98	1.00
Tb ³⁺	−57.3	−838.6	2.403 ± 0.005	8.0–8.2	1.01	1.01
Gd ³⁺	−51.4	−832.7	2.39	8.0–9.9	1.00	1.02
Eu ³⁺	−47.8	−829.1	2.45	8.3	1.06	1.04
Sm ³⁺	−39.4	−820.7	2.448 ± 0.021	8.0–8.9	1.06	1.09
Nd ³⁺	−28.7	−810.0	2.472 ± 0.033	8.0–8.9	1.08	1.13
Pr ³⁺	−20.3	−801.6	2.54	9.2	1.15	1.10
Ce ³⁺	−9.5	−790.8	2.55	7.5	1.16	1.17
La ³⁺	3.6	−777.7	2.528 ± 0.048	8.0–9.1	1.14	1.19
Hf ⁴⁺	−657.0	−1698.1	2.16	8		0.72
Zr ⁴⁺	−615.2	−1656.3	2.19	8		0.75
Ce ⁴⁺	−455.1	−1496.2	2.42	8		1.01
Pu ⁴⁺	−560.2	−1601.3	2.39	8		0.91
U ⁴⁺	−512.4	−1553.5	2.42	9–11		1.01
Th ⁴⁺	−382.2	−1423.3	2.53	9–11	1.14	1.10

^aThe computation refers to Jiang et al.³⁶ using the data from ref 59. ^bComputed by eq 3. ^cIOD and CN from refs 30 and 60 and the references therein for original sources. ^dIonic radii $r_{\text{ion}} = \text{IOD} - r_{\text{water}}$ where r_{water} amounts to ~ 1.4 Å.⁶⁰ ^eAveraged Shannon's revised effective ion radii.⁶¹

presented a systematical development of nonbonded point charge models for monovalent,^{25,26} divalent,^{27–29} and highly charged ions^{30,31} and indicated that the ion–oxygen distance (IOD) and experimental hydration free energy (HFE) for most of the ions cannot be reproduced simultaneously using the traditional 12-6 Lennard-Jones (LJ) potential for vdW interactions. This problem was attributed to the neglect of ion-induced dipole interactions. To solve this, they proposed an additional C_4 item to the 12-6 LJ potential, denoted as 12-6-4 LJ-type, and the C_4 term offered an attractive interaction of cations with water and a repulsive one for anions.^{25,26,28–31} The 12-6-4 LJ-type potential was also used in the nonbonded dummy atom models for the simulations of metalloproteins³⁹ and metal–organic frameworks.³⁷ Other solutions were reported as well, such as the use of a double exponential function for vdW interactions⁴⁰ and of scaled (reduced) ionic charges plus 12-6 LJ potential.⁴¹ The scaled charge strategy by electronic continuum correction^{42–44} described the electronic polarization in nonpolarizable force fields. For polarizable ion models, a different form (buffered 14-7 potential)^{45–47} was adopted to represent the vdW interaction.

Following our previous work on the divalent metal cations,⁴⁸ here we kept the ionic charges fixed and designed nonbonded point charge models with the commonly used 12-6 LJ potential for 18 trivalent metal cations (Al³⁺, Cr³⁺, Fe³⁺, In³⁺, Lu³⁺, Tl³⁺, Tm³⁺, Er³⁺, Y³⁺, Dy³⁺, Tb³⁺, Gd³⁺, Eu³⁺, Sm³⁺, Nd³⁺, Pr³⁺, Ce³⁺, and La³⁺) and 6 tetravalent metal ions (Hf⁴⁺, Zr⁴⁺, Ce⁴⁺, Pu⁴⁺, U⁴⁺, and Th⁴⁺). Eleven models were developed for compatibility with 11 widely used water models, including five three-site water models of TIP3P,⁴⁹ OPC3,⁵⁰ SPC/E,⁵¹ SPC/E_b,⁵² and TIP3P-FB⁵³ and six four-site water models of a99SB-disp,⁵⁴ TIP4P-Ew,⁵⁵ OPC,⁵³ TIP4P/2005,⁵⁶ TIP4P-

D,⁵⁷ and TIP4P-FB.⁵⁸ From an extensive scanning of LJ parameter space, the designed 12-6 LJ models simultaneously reproduced the experimental HFE and IOD of the tested metal ions with good accuracy. The properties that were not targeted in the model development, such as coordination number, diffusion constant, and ion pair interactions, were also evaluated. A metalloprotein of D-xylose isomerase (PDB code: 1XLH) was simulated to demonstrate the effectiveness of our new models. This work provides valuable insights into model developments of metal ions and facilitates computational modeling of highly charged ions in biochemistry and materials science.

COMPUTATIONAL METHODS

Nonbonded Interaction Potential. The nonbonded (NB) interactions (U_{ij}^{NB}) used in this work consist of electrostatic and van der Waals (vdW) contributions; the former is modeled by the Coulomb law and the latter is by 12-6 Lennard-Jones (LJ) potential (eq 1).

$$\begin{aligned}
 U_{ij}^{\text{NB}}(r_{ij}) = & U_{ij}^{\text{ele}}(r_{ij}) + U_{ij}^{\text{vdW}}(r_{ij}) = \frac{1}{4\pi\epsilon_0} \frac{Q_i Q_j}{r_{ij}} \\
 & + \epsilon_{ij} \left(\left(\frac{R_{\text{min}, ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{R_{\text{min}, ij}}{r_{ij}} \right)^6 \right) = \frac{1}{4\pi\epsilon_0} \frac{Q_i Q_j}{r_{ij}} \\
 & + 4\epsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right)
 \end{aligned} \quad (1)$$

where r_{ij} is the atom-pair distance, ϵ_0 is the vacuum permittivity, Q_i and Q_j are partial charges of atoms i and j .

ϵ_{ij} and $R_{\min}$ are the LJ well depth and the atom-pair distance at which the interacting LJ potential reaches the minimum. The vdW part is also described in the form of $\sigma-\epsilon$ (eq 1) and σ is the atom-pair distance ($R_{\min} = 2^{1/6}\sigma$) where the LJ potential amounts to zero. The LJ parameters of $R_{\min,ij}$ and ϵ_{ij} in this work satisfy the Lorentz–Berthelot (LB) mixing rule (eq 2).

$$R_{\min,ij} = \frac{R_{\min,ii} + R_{\min,jj}}{2} = R_{\min,i} + R_{\min,j}, \quad \epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}} \\ = \sqrt{\epsilon_i\epsilon_j} \quad (2)$$

Note that $R_{\min,i}$ is also known as $R_{\min}/2$ in the original Amber force field. For simplification, we used R and ϵ for $R_{\min,i}$ and ϵ_i in units of Å and kcal/mol, respectively.

Experimental Targets. Experimental references of real hydration free energy (HFE), ion–oxygen distance (IOD), and coordination number (CN) in the first hydration shell for the 24 metal cations are given in Table 1, as well as the ionic radii calculated from IOD by Marcus⁶⁰ and Shannon's revised effective ion radii (<http://abulafia.mt.ic.ac.uk/shannon>).⁶¹ The real HFEs of the metal ions were computed by eq 3

$$\Delta G^{*,\text{real}}(\text{M}) = \Delta G^{\text{o,conv}}(\text{M}) + q\Delta G^{\text{o}}(\text{H}^+) - \Delta G^{\text{o}\rightarrow*} \quad (3)$$

where $\Delta G^{\text{o,conv}}$ is conventional hydration free energy (experimental measurable), $\Delta G^{\text{o}}(\text{H}^+)$ is the real hydration free energy of proton (−259.8 kcal/mol) recommended by Hünenberger and Reif,⁶² q is the net charge, and $\Delta G^{\text{o}\rightarrow*}$ is a standard-state correction (1.9 kcal/mol).

General Simulation Setup. Molecular dynamics (MD) simulations were carried out using the GROMACS 2018 software.⁶³ Coulomb interactions were dealt with the particle-mesh Ewald (PME) method with a real-space truncation of 10 Å.^{64,65} The vdW interactions were cut-off at 10 Å with long-range dispersion corrections for energy and pressure. For comparison, the Lennard-Jones PME (LJ-PME) method for a complete (more accurate) long-range dispersion correction was tested as well.^{65,66} If not stated otherwise, the cut-off scheme was used in the simulations. Periodic boundary conditions (PBC) in all directions were used in the simulation. Temperature and pressure were coupled at 298.15 K and 1 bar through the velocity-rescaling⁶⁷ and Parrinello–Rahman algorithms^{68,69} with a coupling time constant of 1 and 5 ps, respectively. The SETTLE method⁷⁰ was used to maintain a rigid water model, and the LINCS algorithm⁷¹ was applied to constraint all chemical bonds (if any). An energy minimization was first implemented to eliminate improper interactions in each system, followed by 100-ps NVT and 400-ps NPT equilibrations. The production simulations were done at the NPT ensemble (if not stated otherwise) for data analysis.

Lennard-Jones Parameter Space. During the simulations of scanning LJ parameter space, the atomic mass of the test metal cations was set to 26.9815 (corresponding to Al^{3+}) and 178.49 (Hf^{4+}) g/mol for trivalent M(III) and tetravalent M(IV) metal cations, respectively. The choice of mass had little effect on the model design.²⁷ Previous reports found that experimental HFE and IOD of highly charged metal cations cannot be reproduced simultaneously in the range of $R \in [0.9, 2.3]$ Å and $\epsilon \in (0, 1)$ kcal/mol.³⁰ Here, we chose R ranging from 0.1 to 2 Å with a step of 0.1 Å (20 points). The value of ϵ ranged from 0.1 to 10 000 kcal/mol, namely, $-\lg(\epsilon) = -4, -3.5, -3.25, -3, -2.5, -2, -1.5, -1, -0.5, 0, 0.5$, and 1 (12 points). For one water model, we examined $20 \times 12 = 240$

different $R-\epsilon$ combinations of 12–6 LJ parameters in total for M(III) and M(IV) ions, respectively.

Hydration Free Energy Calculation. Hydration free energies (HFEs) of the metal ions with different $R-\epsilon$ combinations were computed using a two-stage thermodynamic integration (TI) method following previous work.^{72–75} Electrostatic interactions were decoupled first along with an alchemical reaction coordinate of 25 λ values from 0 to 1 and then 16 λ for decoupling vdW interactions. The HFE was decomposed into two contributions from Coulomb (ΔG_{Coul}) and vdW (ΔG_{vdW}) interactions with water. To avoid the singularities and numerical instabilities in the decoupling, a soft-core potential⁷⁶ was introduced with its λ power parameter setting to 1 and the α_{LJ} parameter to 0.5.⁷⁵ Each λ was performed for 50 ps at 298.15 K and 1 bar, and the first 10 ps was discarded for equilibration. Each simulated system was composed of one metal ion and 600 water molecules in a cubic box with a length of ~ 26.5 Å. The minimum distance between the solute ion and box edge is about 13.2 Å. The HFEs were extracted from TI statistics using the Bennett acceptance ratio (BAR) method⁷⁷ by the GROMACS tool of “gmxbar”.⁶³

The calculated HFEs from typical PME/PBC MD simulations represent the interaction between the ion and the bulk environment and, therefore, belong to the intrinsic energy (ΔG_{intr}). To obtain the physical real free energy of ionic hydration, the surface contribution (ΔG_{surf}) from ions passing through the vacuum/bulk interface is needed to be considered (eq 4).^{74,78,79}

$$\Delta G^{*,\text{real}} = \Delta G_{\text{intr}} + \Delta G_{\text{surf}} = \Delta G_{\text{intr}} + q\phi_{\text{G}} \quad (4)$$

where q is the ionic charge of metal ions (+3 or +4) and ϕ_{G} is the solvent Galvani potential taken from our previous work.⁷² For targeting the real experimental HFE references (Table 1), we choose to compute the real HFEs from MD simulations. The use of real HFEs has an advantage that the computed result is methodology independent.^{62,74,80}

IOD and CN Calculations. Production simulations for one metal ion in 600 water molecules were performed at NPT for 4 ns, and the trajectories were stored every 0.1 ps. The radial distribution functions (RDFs) of the water oxygen atom (OW) around the metal ion were calculated from the last 3-ns trajectories (30 000 frames in total) using the GROMACS tool of “gmxdist” with a bin width of 0.01 Å.⁶³ From the RDFs, we obtained the ion–oxygen distance (IOD) and coordination number (CN) values in the first hydration shell of the metal ions.

Parameter Generation and Validation. The discrete points of HFE and IOD within the LJ parameter space were used to generate the surface of both properties using a two-dimensional (2D) cubic spline interpolation, as did in the refs 36 and 73. Contour lines corresponding to experimental HFE and IOD values of different metal ions were computed and drawn on a two-dimensional graph with $-\lg(\epsilon)$ being the x -axis and R as the y -axis. A Python script was used to do the interpolation and plotting. The intersection point of the HFE and IOD contours was the target parameter combination ($R-\epsilon$), which reproduced both quantities simultaneously. The resulting $R-\epsilon$ combinations were validated by 500-ps TI simulations for each λ and by 4 ns normal MD simulations using the real atomic mass following the same protocol as stated above. If necessary, the LJ parameters were further optimized iteratively to reproduce the experimental HFE

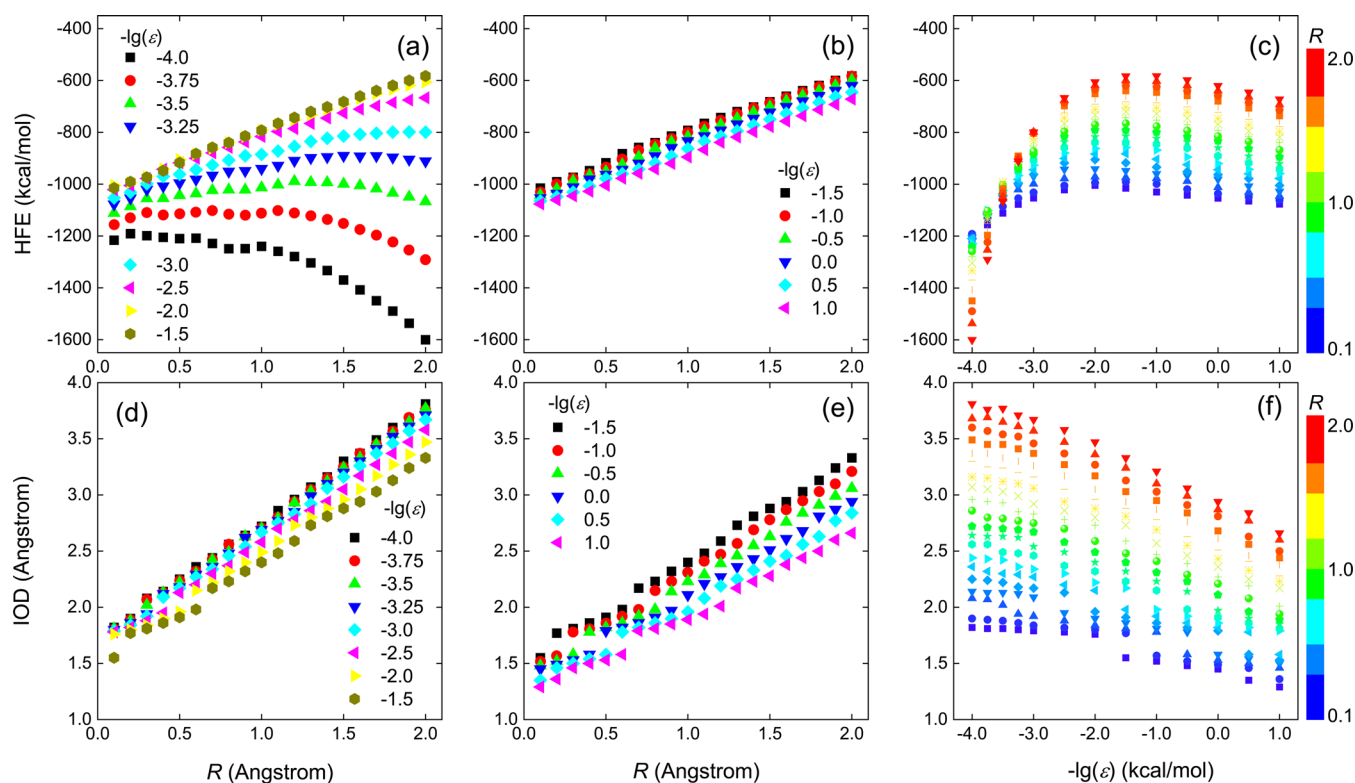


Figure 1. Hydration free energy (HFE) of the test M(III) cation in the TIP3P water model as a function of LJ scanning parameter R for the scanning $-\lg(\epsilon) \leq -1.5$ ($\epsilon \geq 31.6228$ kcal/mol (a)) and $-\lg(\epsilon) \geq -1.5$ (b) and of LJ parameter $-\lg(\epsilon)$ (c). Ion–oxygen distance (IOD) of the test ion as a function of R and $-\lg(\epsilon)$ is presented in panels (d)–(f).

within 1 kcal/mol and the IOD within 0.01 Å. Such optimization led to a tiny change in R and ϵ .

Model Comparison. For comparison with the 12-6 HFE, 12-6 IOD, and 12-6-4 sets by Li and Merz's group,^{30,31} we computed the relative errors of HFE and IOD from experiment and the correlation coefficient between the LJ R parameters of ion models and Shannon's revised effective ion radii.⁶¹ The diffusion constant of Al^{3+} , Cr^{3+} , Fe^{3+} , and Tl^{3+} ions at infinite dilution solution (D_0) were also calculated using eq 5 from 30-ns NVT simulations in a cubic box of 1024 water molecules. The box length is 31.4 Å, with the minimum distance between the ion and the box edge being ~ 15.9 Å.

$$D_0 = \left(D_{\text{PBC}} + \frac{2.83729 k_B T}{6\pi\eta L} \right) \frac{D_{0,W}^{\text{exp}}}{D_{0,W}^{\text{sim}}} \quad (5)$$

where D_{PBC} is the computed diffusion constant in the simulation cubic box with a length of L under periodic boundary conditions (PBC), k_B is the Boltzmann's constant, T is the temperature, η is the simulated solvent viscosity, and $D_{0,W}^{\text{exp}}$ and $D_{0,W}^{\text{sim}}$ are experimental and simulated water diffusion constants, respectively. The simulated η and $D_{0,W}^{\text{sim}}$ for TIP3P and OPC water models were taken from previous work.⁷² Equation 5 corrected the finite-size effects and the diffusion errors in the water model, allowing a meaningful comparison with experiment.^{73,81,82}

Model Test in Salt Solution. To assess the performance of ion-pair interactions, we calculated the osmotic pressure of six trivalent chlorides (AlCl_3 , CeCl_3 , LaCl_3 , NdCl_3 , PrCl_3 , and YCl_3) in aqueous solution with varied molarity concentrations of 0.2, 0.5, 1.0, 1.5, 2.0, 2.5, and 3.0 mol/L, following the protocol by Luo and Roux.⁸³ Three sets of Joung–Cheatham

Cl^- models⁸⁴ compatible with TIP3P, SPC/E, and TIP4P-Ew water models were used in combination with our M(III) models. Osmotic coefficients (φ) were computed using the calculated osmotic pressure (Π) using eq 6 for comparison with experimental observations.^{85,86}

$$\varphi = \frac{\Pi \cdot V_W}{R \cdot T \cdot m \cdot \nu \cdot M_W} \quad (6)$$

where V_W is the partial molal volume of water, R is the gas constant, T is the temperature, m is the molality, ν is the van't Hoff factor (i.e., the total number of cations and anions per formula unit; 4 for AlCl_3), and M_W is the molecular weight of water.^{73,87,88} The density of these salt solutions was extrapolated from the work by Mason⁸⁶ and used for the conversion between molality and molarity.

LJ parameters between Al^{3+} and Cl^- ($R_{\text{Al-Cl}}$ and $\epsilon_{\text{Al-Cl}}$) obtained from the Lorentz–Berthelot (LB) mixing rule failed to reproduce the experimental osmotic coefficients (φ) of an AlCl_3 solution (see the Results and Discussion section). To examine whether an improvement can be obtained by tuning the cation–anion LJ interactions, we modified the LJ σ parameter from 1.2 to 4.5 Å with a step of 0.3 Å, corresponding to $R_{\text{Al-Cl}}/2$ from ca. 0.67 to 2.53 Å; the LJ ϵ parameter was chosen to be 0.1, 0.2, 2.5, 5, 7.5, and 10 kcal/mol. The LB-based $R_{\text{Al-Cl}}$ and $\epsilon_{\text{Al-Cl}}$ were located within the searched parameter space. The resulting 72 different R – ϵ combinations of Al–Cl LJ parameters were then used to compute osmotic coefficients of 3.0 mol/L AlCl_3 solution in which the interactions between the TIP3P-compatible ion models (Al^{3+} or Cl^-) and water molecules were kept unmodified.

Model Test in Protein. To evaluate the performance of our new parameters in biological systems, a metalloprotein of

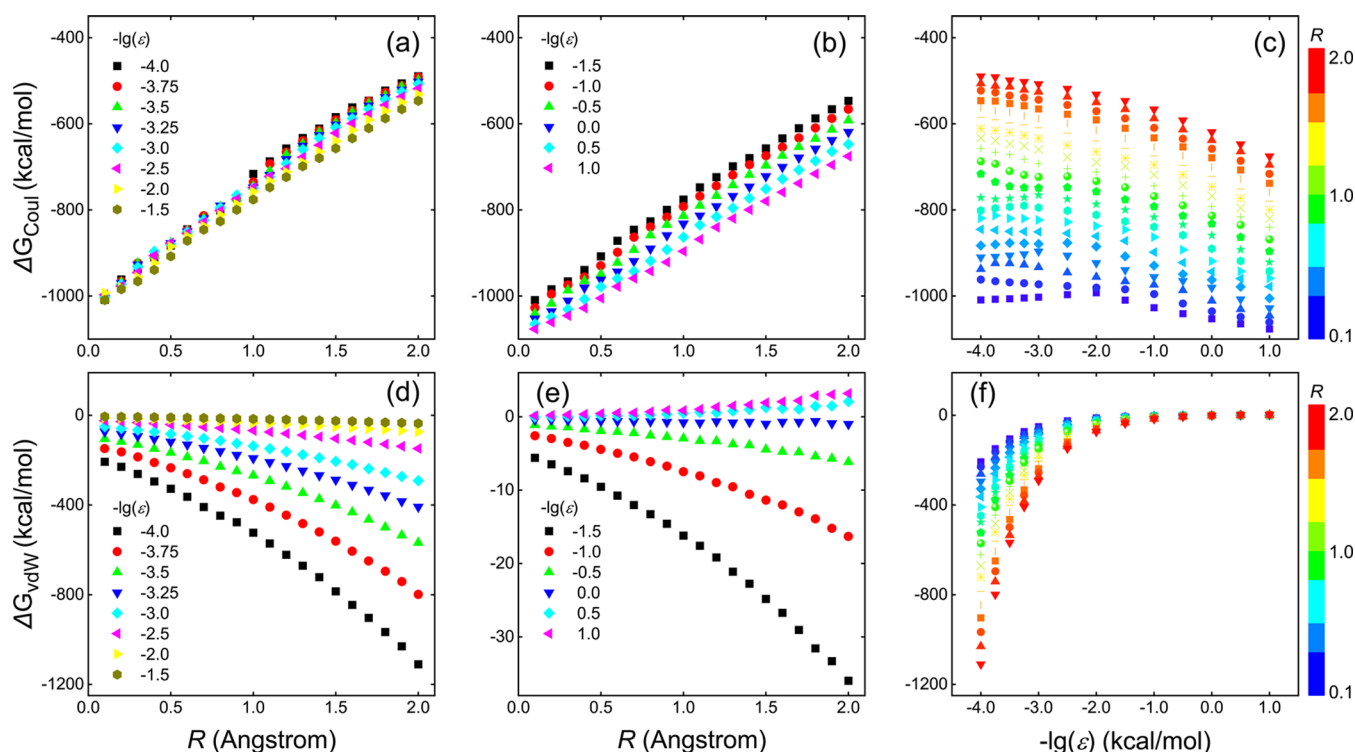


Figure 2. Coulomb (ΔG_{Coul} , (a–c)) and vdW (ΔG_{vdW} , (d–f)) contributions to hydration free energies (HFEs) of the test M(III) cation in the TIP3P water model as a function of LJ scanning parameters of R and ϵ .

D-xylose isomerase (PDB code: 1XLH) with a resolution of 2.5 Å was taken as a test example. Three protein force fields (FF) of Amber14SB,⁸⁹ CHARMM36m,⁹⁰ and a99SB-disp⁵⁴ were used to model the D-xylose isomerase, and the FF-compatible models of TIP3P,⁴⁹ TIPS3P,⁹¹ and a99SB-disp⁵⁴ were chosen to model water molecules, respectively. TIPS3P is a modified TIP3P model with additional LJ sites on hydrogen atoms. Note that we did not develop metal models for use with TIPS3P in this work and for comparison, the FF/water combination of CHARMM36m/TIP3P was tested as well. The crystal water molecules within 5 Å of protein 1XLH were maintained in the preparation of the initial configuration. A cubic box was used with a length of 100 Å in the simulation, and the simulated box contained protein 1XLH (chain A), one Al^{3+} ion, 19 Na^{+} ions, and ca. 31 000 water molecules. During the NVT and NPT equilibrations, a harmonic potential was exerted to constraint the positions of protein backbone atoms. Production trajectories were extended to 50 ns for data analysis and were repeated twice with different initial velocities from a Maxwell distribution at 298.15 K.

RESULTS AND DISCUSSION

Relationship of HFE and IOD with R and ϵ . Due to the correlation between force field parameters, different R – ϵ combinations for the LJ potential can give similar results. Here, we aimed to target simultaneously two properties of hydration free energy (HFE) and ion–oxygen distance (IOD). HFE and IOD discrete points for the test M(III) cation with different R – ϵ combinations in the scanning LJ space for the TIP3P water model are presented in Figure 1, indicating a relationship with R and ϵ .

For $-\lg(\epsilon) \leq -3.25$, the HFE of the test metal ion increases with the increasing R at first, while it shows a decreasing trend for large R (Figure 1a). This is caused by the increase in the

vdW interaction as R increases, although a larger R corresponds to a smaller electrostatic interaction due to the large ion–water distance. For a large ϵ , the increase of the vdW interaction is much larger than the decrease of the electrostatic interaction, as revealed by the changes in Coulomb (ΔG_{Coul} , Figure 2a–c) and vdW (ΔG_{vdW} , Figure 2d–f) components of the HFEs. When $-\lg(\epsilon) \geq -3.0$, that is, $\epsilon \leq 1000$ kcal/mol, the HFEs increase monotonically as R increases (Figure 1a,b), a landscape similar to ΔG_{Coul} (Figure 2a–c). For $-\lg(\epsilon) \leq -1.5$, a larger ϵ gives a more negative HFE for a certain R . When $-\lg(\epsilon)$ is greater than -1.5 , however, increasing ϵ tends to give a less negative HFE, that is, a weaker interaction between the metal and water molecules (Figure 1a–c). This can be ascribed to the fact that the increase of the vdW interaction is much smaller when the potential well depth (ϵ) becomes smaller. For a small ϵ , ΔG_{vdW} contributes little to the HFE (Figure 2e); ΔG_{Coul} is the decisive factor and appears to increase (less negative) with the increasing ϵ (Figure 2a–c).

IOD increases monotonically with the increasing R as well, and as expected, a bigger R produces a larger IOD. Our scanning LJ space appears impossible to locate an IOD between 1.6 and 1.8 approximately, as indicated by an IOD jump (Figure 1d–f). It is not a problem because our targeted IOD values (Table 1) are not in this region. Such a jump was observed as well in the development of divalent metal cations.⁴⁸ For a large R , IOD seems to decrease linearly with the increase in $-\lg(\epsilon)$; for a small R , IOD appears not sensitive to the change in ϵ (Figure 1f). Similar findings are observed for the test M(IV) cation in the TIP3P water model (Figures S1 and S2 in the Supporting Information, SI) as well as the other water models; see the compressed ZIP files for HFE and IOD discrete points with the 11 water models in the SI.

Model Generation and Validation. As did in our previous work for dummy atom models of divalent metal

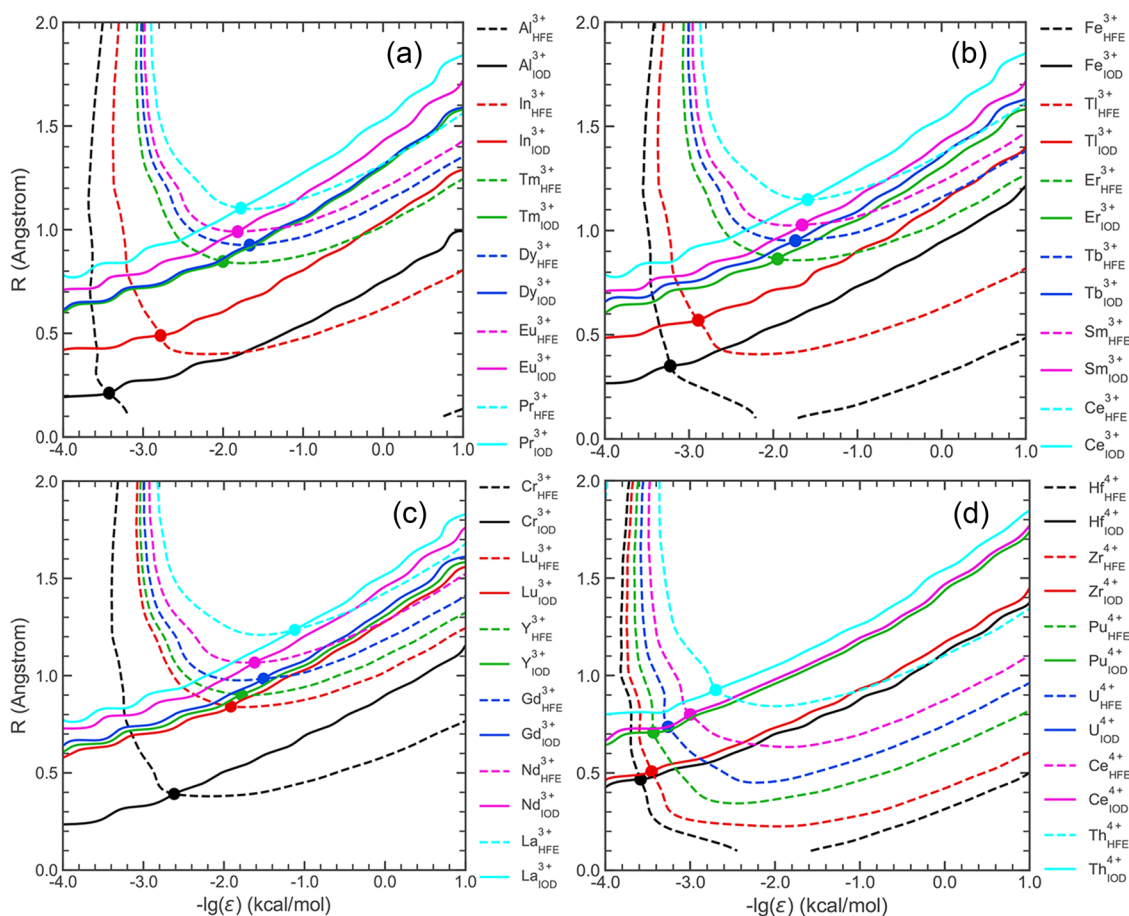


Figure 3. Contour plots of hydration free energy (HFE, dashed line) and the ion–oxygen distance (IOD, solid line) for (a) Al^{3+} , In^{3+} , Tm^{3+} , Dy^{3+} , Eu^{3+} , and Pr^{3+} ions, for (b) Fe^{3+} , Tl^{3+} , Er^{3+} , Tb^{3+} , Sm^{3+} , and Ce^{3+} ions, for (c) Cr^{3+} , Lu^{3+} , Y^{3+} , Gd^{3+} , Nd^{3+} , and La^{3+} ions, and for (d) Hf^{4+} , Zr^{4+} , Pu^{4+} , U^{4+} , Ce^{4+} , and Th^{4+} ions in the parameter space of $R \in [0, 2.0]$ Å and $-\lg(\epsilon) \in [-4.0, 1.0]$ kcal/mol in the TIP3P water. Experimental HFE and IOD references in Table 1 are used to construct the corresponding contour lines. The intersection points of HFE and IOD curves are the target Lennard-Jones 12-6 parameters for these metal cations.

ions,^{36,73} we used the HFE and IOD discrete points in the scanned parameter space to construct the HFE and IOD surfaces and then plotted the contour lines corresponding to experimental references in an x – y diagram. The contour curves for the 24 metal cations in the TIP3P water model are given in Figure 3. We can see that HFE and IOD curves tend to be parallel in some cases for $-\lg(\epsilon) > 0$ ($\epsilon < 1$ kcal/mol), implying that it is impossible to reproduce these two quantities simultaneously using a small ϵ . This is similar to the observed deficiency in developing 12-6 LJ point charge models for mono-, di-, tri-, and tetravalent ions by Li and Merz's group.^{25–27,29–31} When the search LJ space is extended to large ϵ , the HFE contours behave like an upward parabola, facilitating an intersection with the IOD curves.

To obtain a certain HFE value with a large ϵ , one can choose to increase the LJ parameters of R and ϵ or decrease both parameters, as indicated by the left part of HFE contours (Figure 3). The vdW interaction with a large ϵ makes a great contribution to the HFE, and the decrease of electrostatic interactions caused by the increasing R can be largely compensated for by the increase of vdW interactions (Figure 2d,f). The former choice yields small electrostatic interactions and relatively large vdW interactions, while the latter corresponds to large electrostatic interactions and relatively small vdW interactions (Figure 2a,d). However, the case is different for a small ϵ ; the vdW interaction (ΔG_{vdW}) is small,

and ΔG_{Coul} contributes greatly to the HFE (Figure 2b,e). For a certain HFE, one needs a smaller R and a larger ϵ or a larger R and a smaller ϵ , similar to the change in ΔG_{Coul} with R and ϵ (Figure 2b,c). This is depicted as well in the right part of HFE contours in Figure 3.

The intersection points between HFE and IOD contours are the designed 12-6 LJ parameters that simultaneously reproduce both properties for the metal cations. Table 2 listed the final optimized LJ parameters for the TIP3P water model as well as the simulated HFE, IOD, and CN values in the first hydration shell for the tested 24 metal ions. The root-mean-square error (RMSE), mean absolute error (MAE), and mean signed error (MSE) from experimental real HFEs for the designed metal models are 0.5, 0.4, and -0.3 kcal/mol, respectively. The IOD values in the first hydration shell are reproduced within 0.01 Å on average. The two methods of handling vdW interactions (cut-off and LJ-PME, as stated in the Computational Methods section) give almost identical predictions on the IOD, while LJ-PME yields a less negative HFE by ~ 1 kcal/mol than the cut-off method (Figure S3, SI). The CN values of Al^{3+} , Cr^{3+} , Gd^{3+} , Sm^{3+} , Nd^{3+} , and U^{4+} are reproduced accurately, while for other metals, an overestimation by 1–4 is observed. Li et al. chose a small LJ well depth ($\epsilon < 1$ kcal/mol)³⁰ and found that the 12-6 LJ potential failed to reproduce HFE and IOD simultaneously of the same M(III) and M(IV) cations. They then added an attractive C_4 term to the 12-6 LJ potential,

Table 2. Optimized Parameters and Simulated Real Hydration Free Energy (HFE, kcal/mol), Ion–Oxygen Distance (IOD, Å), and Coordination Number (CN) for M(III) and M(IV) Ions with the TIP3P Water Model^a

metal	R	ϵ	HFE	IOD	CN
Al ³⁺	0.2120	2666.2447	−1108.1 ± 0.1	1.88	6.0
Cr ³⁺	0.3928	415.1452	−984.5 ± 0.1	1.96	6.0
Fe ³⁺	0.3503	1710.8030	−1045.1 ± 0.2	2.04	7.2
In ³⁺	0.4894	605.8987	−977.1 ± 0.3	2.15	8.0
Lu ³⁺	0.8412	82.1297	−865.9 ± 0.2	2.34	9.0
Ti ³⁺	0.5756	834.0651	−975.4 ± 0.2	2.22	8.1
Tm ³⁺	0.8466	100.4847	−866.2 ± 0.1	2.36	9.0
Er ³⁺	0.8639	88.7360	−861.3 ± 0.1	2.36	9.0
Y ³⁺	0.9015	60.7435	−851.1 ± 0.1	2.37	9.0
Dy ³⁺	0.9256	46.5479	−845.1 ± 0.2	2.37	9.0
Tb ³⁺	0.9512	54.5507	−839.0 ± 0.1	2.40	9.0
Gd ³⁺	0.9842	32.2478	−831.8 ± 0.1	2.39	9.0
Eu ³⁺	0.9903	65.8415	−829.5 ± 0.1	2.44	9.0
Sm ³⁺	1.0266	45.4255	−820.6 ± 0.1	2.45	9.0
Nd ³⁺	1.0674	41.7061	−810.0 ± 0.2	2.47	9.3
Pr ³⁺	1.1017	57.2532	−802.8 ± 0.2	2.55	10.0
Ce ³⁺	1.1483	39.1021	−791.4 ± 0.1	2.56	10.0
La ³⁺	1.2345	13.2221	−777.8 ± 0.1	2.54	9.9
Hf ⁴⁺	0.4671	3926.4494	−1698.4 ± 0.2	2.16	8.8
Zr ⁴⁺	0.5073	2844.4611	−1656.7 ± 0.3	2.19	9.0
Ce ⁴⁺	0.8001	999.3095	−1496.6 ± 0.3	2.41	10.0
Pu ⁴⁺	0.7077	2644.8436	−1601.7 ± 0.3	2.38	10.0
U ⁴⁺	0.7667	1907.2165	−1553.9 ± 0.4	2.41	10.0
Th ⁴⁺	0.9035	426.5795	−1424.3 ± 0.2	2.54	12.0
RMSE			0.5	0.01	
MAE			0.4	0.01	
MSE			−0.3	0.00	

^aRoot-mean-square error (RMSE), mean absolute error (MAE), and mean signed error (MSE) from the experiment were computed to evaluate the model performance.

thereby resulting in more negative HFEs and solving the problem. The LJ well depth (ϵ) in our designed models ranges from 10 to 4000 kcal/mol for the TIP3P water model (Table 2), and we solve the problem by increasing the vdW interactions of metal cations with water to a large extent. This indicates that the C_4 coefficient can likely be merged into the C_{12} and C_6 terms by increasing the LJ well depth somehow.

After parameter generation and (if necessary) subsequent optimization, the final optimized LJ parameters are tabulated in Table 3 for use with the other water models of OPC3, SPC/E, SPC/E_b, and TIP3P-FB, in Table 4 for a99SB-disp, TIP4P-Ew, and in Table 5 for OPC, TIP4P/2005, TIP4P-D, and TIP4P-FB. The root-mean-square error (RMSE), mean absolute error (MAE), and mean signed error (MSE) from the experiment are given in Table 4, S1, and S2 (SI). These metal models reproduce the experimental real HFEs and IOD within 1 kcal/mol and 0.01 Å on average for all of the tri- and tetravalent metal cations, respectively, and the coordination number (CN) is predicted reasonably in most cases.

Model Comparison. Relative errors of the simulated HFE and IOD values from experiment (Table 1) for the 24 metal cations in TIP3P water are shown in Figure 4, along with a comparison with the 12-6-4 LJ set by Li and Merz's group.³⁰ The comparisons of the two LJ-type models for SPC/E, TIP4P-Ew, OPC3, OPC, TIP3P-FB, and TIP4P-FB water models are given in Figures S4–S9 in the SI, respectively.

Either 12-6 or 12-6-4 LJ-type potential for the metal ions with the seven water models are able to reproduce HFE and IOD within the relative errors of 0.3% and 0.8%, respectively, except for the 12-6-4 LJ-type model of Pr³⁺ in TIP4P-FB displaying a large HFE error of 2.6% (Figure S9). In terms of HFE and IOD, the accuracy of our new 12-6 LJ models is equivalent to or better than the 12-6-4 LJ-type models.^{30,31}

Ionic radius is one of the most important parameters for the prediction of the physical properties of metal ions. Similar to previous work on the divalent metal cations,^{40,48} we chose Shannon's revised effective ion radii (Table 1) as a reference and compared it with the LJ R parameters of different ion models. Li and Merz^{30,31} found that the 12-6 LJ potential failed to reproduce HFE and IOD simultaneously (actually, it worked when enlarging the LJ scanning space as did in this work) and developed 12-6 HFE and IOD sets targeting only one property. They also designed a set for reproducing both the properties using the 12-6-4 LJ-type potential. Li and Merz's HFE, IOD, and 12-6-4 sets for the TIP3P water model largely overestimate the ion radii compared to Shannon's radii data, as indicated by a positive mean signed error (MSE) of 0.4–0.7 Å (Figure 5). The R parameters of our models for TIP3P water tend to give a small underestimation by −0.14 Å (Figure 5). For all of the water models, the RMSE of Li and Merz's models amounts to 0.4–0.7 Å, larger than our models (~0.2 Å). Their models overestimated the ion radii by 0.3–0.7 Å, while our models yield an underestimation by 0.1–0.2 Å (Table S3).

For Li and Merz's three sets and our models of the M(III) metal ions, the Pearson correlation coefficient (PCC) and Spearman rank correlation coefficient (SRCC) are larger than 0.97 (Table S3), indicating that the relative ionic radii are well reproduced. For the M(IV) metal ions, Li and Merz's HFE and 12-6-4 sets yield a PCC value ≤0.94 and a SRCC value ≤0.90, smaller than that of our models (≥0.99). Li and Merz's IOD set targeting the ion–oxygen distance (IOD) has a better performance in PCC and SRCC than their HFE and 12-6-4 sets (Table S3). For the M(III) and M(IV) ions, our models show high PCC and SRCC values than the three sets by Li and Merz's group on average (Figure 5 and Table S3). These findings indicate that the LJ R parameters of our designed 12-6 LJ models have a stronger correlation with Shannon's effective ionic radii than Li and Merz's three sets.

Lennard-Jones (LJ) potential functions were initially proposed to describe nonpolar interactions between for instance noble gases. The 12-6 LJ R parameter (eq 1) is the known vdW radius. Ionic radii are the radii of ions obtained from crystal structures where the interaction is mainly the ionic bond between the cation and the anion. In aqueous solution, metal cations interact with water molecules via coordination bonds that are described by electrostatic and vdW interactions in point charge models. Ionic radii are therefore different from vdW radii; the former is typically smaller than the latter.^{61,92} Based on the experimentally determined ion–oxygen distance (IOD), Marcus compiled a set of ionic radii (r_{ion}) by $r_{\text{ion}} = \text{IOD} - r_{\text{water}}$ where r_{water} amounts to ~1.4 Å,⁶⁰ and this set is very close to Shannon's ionic radii (Table 1).⁶¹

For a physical model, the LJ R parameter is expected to be close to the vdW radius. To reproduce the IOD and HFE properties of highly charged ions simultaneously, however, one needs a relatively small 12-6 LJ R parameter, which is close to the ionic radius (Figure 5). This appears to be somewhat not physically meaningful. When the R parameters increased being close to the vdW radii, the 12-6 LJ model would fail to

Table 3. Optimized Parameters for M(III) and M(IV) Metal Ions with Three-Site Water Models of OPC3, SPC/E, SPC/E_b, and TIP3P-FB

metal	OPC3		SPC/E		SPC/E _b		TIP3P-FB	
	R	ϵ	R	ϵ	R	ϵ	R	ϵ
Al ³⁺	0.2243	1678.8040	0.2179	2335.6082	0.2200	2238.2057	0.2043	2162.2206
Cr ³⁺	0.4436	161.7335	0.4059	315.6458	0.4152	297.5089	0.4068	311.9608
Fe ³⁺	0.3648	1192.6143	0.3505	1580.5201	0.3545	1533.9104	0.3437	1589.6445
In ³⁺	0.4981	352.9393	0.4869	531.0067	0.4826	510.7402	0.4839	533.2121
Lu ³⁺	0.9002	41.0866	0.8519	72.2271	0.8594	63.1830	0.8485	72.8115
Tl ³⁺	0.5915	550.9345	0.5767	738.7543	0.5894	751.9691	0.5865	793.5969
Tm ³⁺	0.8990	53.2599	0.8563	85.6249	0.8624	77.4462	0.8529	88.9816
Er ³⁺	0.9195	43.2813	0.8734	74.7997	0.8806	65.7204	0.8697	75.0931
Y ³⁺	0.9757	23.9828	0.9156	46.7197	0.9234	42.1114	0.9102	43.3112
Dy ³⁺	1.0048	19.3108	0.9460	33.6589	0.9500	34.5223	0.9412	30.8674
Tb ³⁺	1.0238	23.3776	0.9710	39.7283	0.9739	41.5050	0.9670	34.4350
Gd ³⁺	1.0948	8.5409	1.0068	22.3409	1.0021	27.6376	0.9987	23.1420
Eu ³⁺	1.0525	31.8273	1.0091	55.5265	1.0121	56.0661	1.0060	45.6562
Sm ³⁺	1.1178	13.6458	1.0476	35.9501	1.0467	38.7258	1.0417	34.9140
Nd ³⁺	1.1646	12.4710	1.0901	31.7687	1.0901	32.4788	1.0820	35.2371
Pr ³⁺	1.1752	23.3131	1.1145	49.8884	1.1209	47.1954	1.1110	51.2271
Ce ³⁺	1.2307	15.4739	1.1641	31.5355	1.1723	27.9190	1.1629	27.9319
La ³⁺	1.3516	4.5113	1.2710	8.1527	1.2570	11.1506	1.2581	9.8220
Hf ⁴⁺	0.4774	2876.0736	0.4618	3480.1663	0.4611	3346.5707	0.4711	3624.0986
Zr ⁴⁺	0.5243	2061.1045	0.5049	2521.1576	0.5113	2510.7299	0.4969	2511.3081
Ce ⁴⁺	0.8143	636.7955	0.8033	889.2011	0.8044	851.5301	0.8029	896.1898
Pu ⁴⁺	0.7149	1984.2665	0.7035	2384.5147	0.7044	2314.1953	0.6951	2380.6743
U ⁴⁺	0.7850	1412.8628	0.7680	1731.0116	0.7766	1713.1682	0.7645	1709.2280
Th ⁴⁺	0.9678	366.7874	0.9297	443.4044	0.9560	452.3759	0.9341	460.6807

Table 4. Optimized Parameters and Simulated Real Hydration Free Energy (HFE, kcal/mol), Ion–Oxygen Distance (IOD, Å), and Coordination Number (CN) for M(III) and M(IV) Metal Ions with a99SB-disp and TIP4P-Ew Water Models

metal	a99SB-disp					TIP4P-Ew				
	R	ϵ	HFE	IOD	CN	R	ϵ	HFE	IOD	CN
Al ³⁺	0.1565	2668.0871	−1108.3 ± 0.2	1.88	6.0	0.1928	4087.8975	−1107.2 ± 0.2	1.89	6.0
Cr ³⁺	0.3334	534.1952	−985.3 ± 0.2	1.96	6.0	0.3336	835.4106	−983.7 ± 0.2	1.96	6.0
Fe ³⁺	0.3446	2144.8651	−1045.9 ± 0.3	2.02	6.0	0.3021	2652.1617	−1044.6 ± 0.4	2.03	6.9
In ³⁺	0.4408	936.9146	−978.3 ± 0.3	2.15	7.5	0.4534	1122.0185	−978.1 ± 0.3	2.16	8.0
Lu ³⁺	0.7556	162.6672	−866.5 ± 0.2	2.34	8.2	0.7475	203.8449	−867.1 ± 0.3	2.34	9.0
Tl ³⁺	0.5165	944.4957	−974.9 ± 0.4	2.23	8.0	0.5491	1303.7671	−974.4 ± 0.2	2.23	8.1
Tm ³⁺	0.7704	183.4003	−867.0 ± 0.2	2.36	8.8	0.7820	257.8101	−866.1 ± 0.2	2.37	9.0
Er ³⁺	0.7802	162.8921	−861.8 ± 0.1	2.37	9.0	0.7907	229.1395	−861.8 ± 0.2	2.37	9.0
Y ³⁺	0.8023	122.3770	−851.6 ± 0.1	2.37	9.0	0.8050	156.2068	−851.1 ± 0.3	2.37	9.0
Dy ³⁺	0.8196	99.2888	−845.0 ± 0.2	2.36	9.1	0.8186	125.9215	−845.2 ± 0.2	2.36	9.0
Tb ³⁺	0.8554	104.1118	−839.5 ± 0.2	2.41	9.0	0.8543	131.2502	−838.7 ± 0.2	2.40	9.0
Gd ³⁺	0.8674	75.9277	−833.0 ± 0.2	2.39	9.0	0.8687	100.3460	−832.1 ± 0.1	2.39	9.0
Eu ³⁺	0.9106	102.5416	−828.8 ± 0.2	2.46	9.0	0.9066	144.6105	−828.2 ± 0.2	2.44	9.0
Sm ³⁺	0.9203	73.3499	−821.7 ± 0.1	2.44	9.0	0.9312	116.2519	−819.8 ± 0.2	2.44	9.0
Nd ³⁺	0.9709	63.9882	−810.2 ± 0.1	2.47	9.0	0.9689	102.0234	−810.0 ± 0.1	2.46	9.1
Pr ³⁺	1.0264	86.4569	−801.0 ± 0.2	2.53	9.1	1.0180	119.0066	−801.2 ± 0.2	2.55	10.0
Ce ³⁺	1.0579	60.5620	−789.9 ± 0.2	2.54	9.0	1.0539	101.7185	−791.7 ± 0.2	2.56	10.0
La ³⁺	1.0971	31.9669	−778.5 ± 0.2	2.53	9.0	1.0853	44.7301	−778.7 ± 0.1	2.53	9.6
Hf ⁴⁺	0.4234	4140.9501	−1697.5 ± 0.2	2.16	8.0	0.4391	5550.0897	−1697.1 ± 0.3	2.15	8.2
Zr ⁴⁺	0.4714	3181.2661	−1656.1 ± 0.3	2.19	8.0	0.4761	4228.6331	−1656.4 ± 0.2	2.19	8.9
Ce ⁴⁺	0.7574	1177.6060	−1496.4 ± 0.2	2.42	9.6	0.7557	1538.1546	−1495.7 ± 0.2	2.42	10.0
Pu ⁴⁺	0.6848	2747.8942	−1602.0 ± 0.4	2.38	9.5	0.6917	3650.0600	−1601.5 ± 0.4	2.40	10.0
U ⁴⁺	0.7218	1927.0811	−1552.8 ± 0.3	2.42	9.9	0.7346	2642.4088	−1553.2 ± 0.5	2.42	10.0
Th ⁴⁺	0.8963	673.2867	−1423.1 ± 0.4	2.52	10.0	0.8667	819.9739	−1423.1 ± 0.2	2.53	11.2
RMSE			0.7	0.01				0.6	0.01	
MAE			0.6	0.01				0.5	0.01	
MSE			−0.3	0.00				0.1	0.00	

Table 5. Optimized Parameters of M(III) and M(IV) Metal Ions for Four-Site Water Models of OPC, TIP4P/2005, TIP4P-D, and TIP4P-FB

metal	OPC		TIP4P/2005		TIP4P-D		TIP4P-FB	
	R	ϵ	R	ϵ	R	ϵ	R	ϵ
Al ³⁺	0.1756	2032.3570	0.1798	4048.5546	0.1605	2894.6762	0.1988	2953.9286
Cr ³⁺	0.3811	363.5800	0.3233	966.0509	0.3304	602.4208	0.3596	524.8074
Fe ³⁺	0.3565	1655.6733	0.3142	2855.6172	0.3338	2260.4765	0.3344	2065.8558
In ³⁺	0.4700	646.2492	0.4389	1245.3746	0.4377	986.7338	0.4636	809.6550
Lu ³⁺	0.8127	97.4092	0.7435	246.4336	0.7533	174.7029	0.7864	123.3105
Tl ³⁺	0.5578	759.4514	0.5405	1355.8137	0.5158	1004.3845	0.5585	959.8425
Tm ³⁺	0.8221	112.3311	0.7581	270.0847	0.7668	196.1553	0.7977	142.7579
Er ³⁺	0.8358	97.0063	0.7694	245.1884	0.7767	175.0653	0.8103	122.4616
Y ³⁺	0.8687	65.4787	0.7751	174.5823	0.7990	130.1368	0.8427	86.0994
Dy ³⁺	0.8898	56.3378	0.8035	157.2896	0.8156	110.8664	0.8642	71.7960
Tb ³⁺	0.9186	60.0620	0.8345	156.9278	0.8512	111.7892	0.8962	80.2417
Gd ³⁺	0.9317	39.9852	0.8394	118.7682	0.8586	79.5426	0.9095	57.2137
Eu ³⁺	0.9627	63.3578	0.8828	158.4528	0.9053	111.5578	0.9465	104.4720
Sm ³⁺	0.9931	38.1505	0.9023	123.8226	0.9242	82.0540	0.9672	76.4892
Nd ³⁺	1.0410	32.9837	0.9446	111.9438	0.9654	71.5319	1.0089	59.6348
Pr ³⁺	1.0821	55.3350	0.9986	143.1528	1.0217	99.2659	1.0544	79.7995
Ce ³⁺	1.1209	38.9762	1.0344	109.5973	1.0555	70.0487	1.0932	57.0821
La ³⁺	1.2008	12.9688	1.0647	61.4328	1.0936	33.2660	1.1473	25.1594
Hf ⁴⁺	0.4367	3388.4415	0.4360	5747.7782	0.4208	4467.8646	0.4412	4257.9445
Zr ⁴⁺	0.4988	2609.1557	0.4683	4388.3373	0.4780	3488.1889	0.4869	3207.7462
Ce ⁴⁺	0.7823	907.6115	0.7527	1623.6783	0.7483	1249.9712	0.7830	1163.8580
Pu ⁴⁺	0.7010	2329.1636	0.6920	3659.3167	0.6821	2934.2695	0.6940	2866.1571
U ⁴⁺	0.7429	1604.3533	0.7153	2618.1830	0.7163	2070.1413	0.7375	2011.8704
Th ⁴⁺	0.9182	483.9494	0.8618	910.1228	0.8976	734.8522	0.9030	642.3918

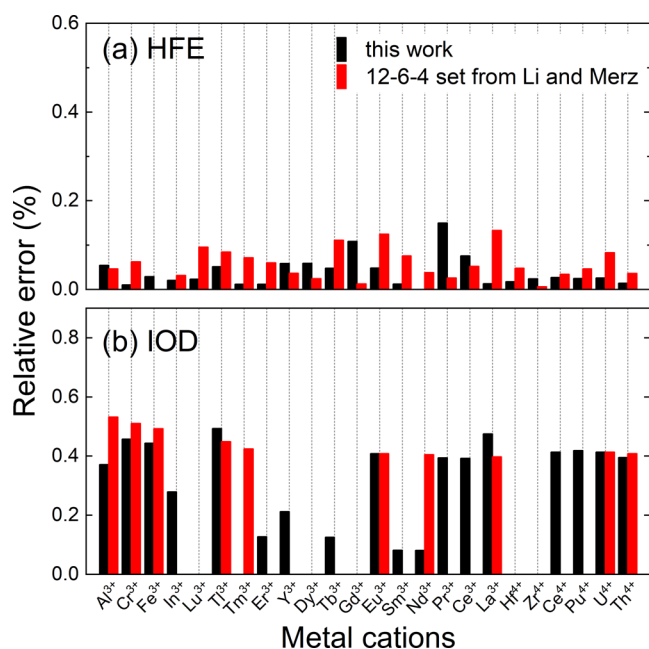


Figure 4. Relative errors of the calculated HFE (a) and IOD (b) values from experiment (Table 1) for the 24 metal cations in TIP3P water (Table 2) using the 12-6 LJ model in this work (black) and the 12-6-4 set (red) by Li and Merz's group.³⁰

reproduce the two properties of ions simultaneously, as confirmed by Li and Merz's mono-,²⁶ di-,^{27,29} and trivalent ion models.³⁰ For instance, Li and Merz's 12-6 HFE set of divalent ions in TIP3P water underestimated the ion–oxygen distance (IOD) by ~ 0.3 Å, while their 12-6 IOD set yielded more positive HFEs by ~ 50 kcal/mol. An attractive C_4 term

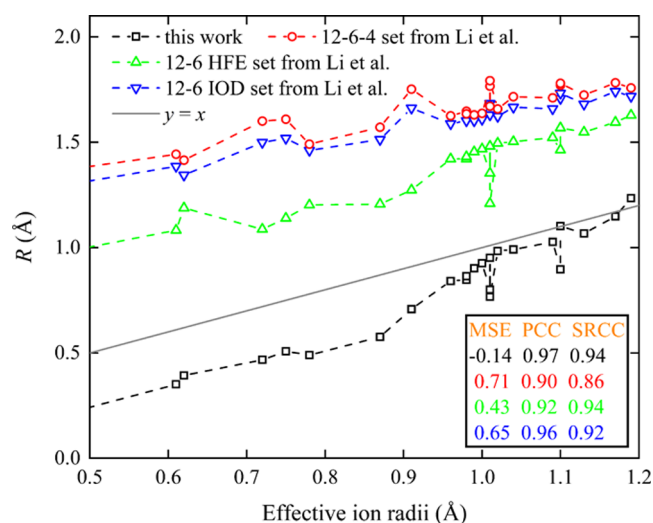


Figure 5. Correlation of Shannon's revised effective ion radii with Lennard-Jones R parameters of four point charge models in the TIP3P water model for the M(III) and M(IV) metal cations, including the 12-6 HFE, 12-6 IOD, and 12-6-4 sets from Li and Merz's group and our designed models. Mean signed error (MSE) from the Shannon's radii, Pearson correlation coefficient (PCC), and Spearman rank correlation coefficient (SRCC) are listed at the bottom right for comparison. The solid gray line indicates a perfect match ($y = x$).

was thus proposed to solve the problem. However, the case is different for the anions. Using LJ R parameters close to the vdW radii of anion ions, Li and Merz's 12-6 IOD set underestimated the HFEs (more negative) and a repulsive C_4 term was needed.^{25,26} A variety of water models use an LJ R

Table 6. Simulated Diffusion Constants (D_0) of Several M(III) Ions at Infinite Aqueous Solution^a

model	water	vdW-type ^b	Al ³⁺	Cr ³⁺	Fe ³⁺	Tl ³⁺
this work	TIP3P	cut-off	0.843 ± 0.083	0.819 ± 0.067	0.725 ± 0.060	0.705 ± 0.055
this work	TIP3P	LJ-PME	0.780 ± 0.075	0.742 ± 0.074	0.732 ± 0.068	0.663 ± 0.061
this work	TIP4P/2005	cut-off	0.907 ± 0.030	0.826 ± 0.036	0.747 ± 0.023	0.731 ± 0.025
this work	TIP4P/2005	LJ-PME	0.835 ± 0.029	0.826 ± 0.032	0.712 ± 0.028	0.750 ± 0.024
this work	OPC	cut-off	0.776 ± 0.027	0.797 ± 0.029	0.724 ± 0.046	0.639 ± 0.025
12-6-4 set	OPC	cut-off	0.710 ± 0.061	0.696 ± 0.103	0.687 ± 0.089	
exp ^c			0.541	0.595	0.604	0.61

^a D_0 ($\times 10^{-5}$ cm²/s) was computed by eq 5; the results for the 12-6-4 set were taken from Li et al.³¹ and the diffusion errors in the water model were corrected (eq 5). ^bThe type of handling vdW interactions by the cut-off scheme with a dispersion correction of energy and pressure and by the Lennard-Jones PME (LJ-PME) scheme.^{65,66} ^cExperimental observations were taken from ref 98.

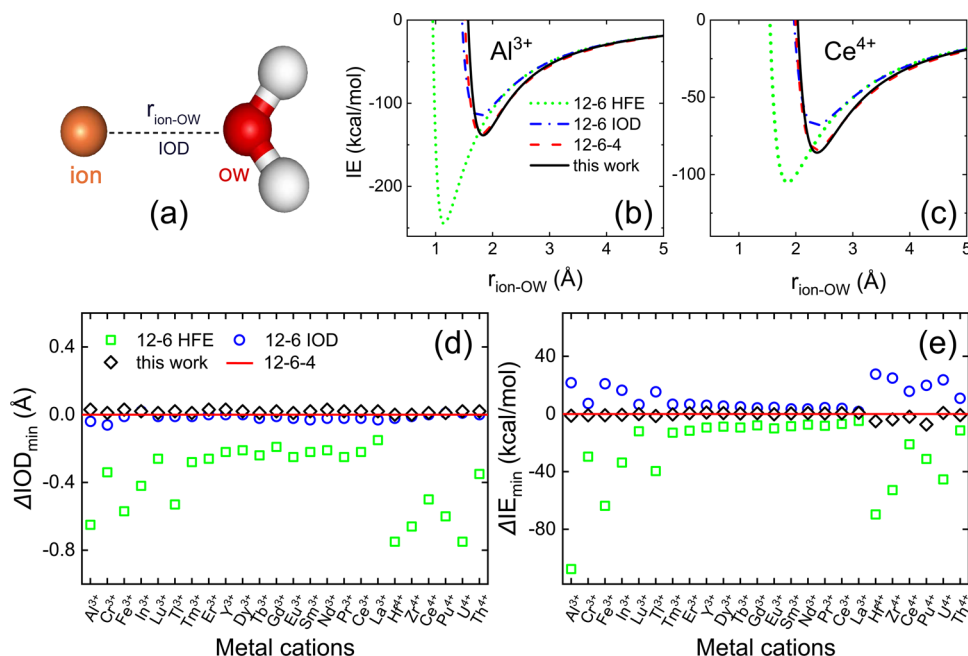


Figure 6. (a) Illustration for computing nonbonded interaction energies between M^{3+}/M^{4+} ions and one water molecule as a function of ion–OW distance (IOD). (b) Interaction energy (IE) between Al^{3+} and TIP3P water for different models from this work and the 12-6 HFE, 12-6 IOD, and 12-6-4 sets by Li and Merz's group.³⁰ (c) IE between Ce^{4+} and TIP3P water. (d) and (e) show the IOD and IE differences from the 12-6-4 LJ set, respectively. IOD_{min} is the distance where IE reaches its minimum (IE_{min}). The solid red lines indicate a value of zero for ΔIOD and ΔIE in panels (d) and (e), respectively, when using the 12-6-4 set as a reference.

parameter of 1.7–1.8 Å to reproduce the experimental oxygen–oxygen distance of ~ 2.8 Å.^{49,93,94} If a model with a water radius of $R \sim 1.4$ Å could reproduce the experimental properties of pure water accurately, using the ionic radii as the LJ R parameters of ion models would make more sense. If not, a better way is to improve the physical model and parametrize a more accurate model then.

Diffusion constants (D_0) of Al^{3+} , Cr^{3+} , Fe^{3+} , and Tl^{3+} ions in TIP3P, TIP4P/2005, and OPC water at infinite dilution were computed with the corrections of finite-size effects and of diffusion errors in the water model (eq 5). The choice of methods (cut-off or LJ-PME) for vdW interactions has little influence on the properties of pure water such as oxygen–oxygen distance, density, interaction energy (IE), diffusion constant, and viscosity (Table S5, SI) and on the diffusion constants of the ions (Table 6). Nonequilibrium simulations of 1000 water molecules with an external force^{95,96} in a cubic box with a length of ~ 31 Å were also attempted to calculate the water viscosity, and the results (Table S5) are almost identical to our previous work using equilibrium simulations.⁷² The former nonequilibrium method is much cheaper than the

latter, but it has the drawback that only one property can be obtained for each simulation.⁹⁷

The calculated D_0 of ions overestimates the experimental observations by 5–68%, and the prediction with OPC-compatible ion models is much close to the experiment compared to that of TIP4P/2005 and TIP3P-compatible models. Compared to the 12-6-4 LJ-type models, our models give a slightly faster diffusivity of the tested M(III) ions (Table 6). The TIP4P/2005 model reproduces the water diffusion constant accurately (Table S5), which eliminates the diffusion error in the water model when computing diffusion constants of the ions (eq 5). However, using this model still leads to an overestimation of the ion diffusion, and this may be due to the TIP4P/2005 model underestimating the water viscosity slightly (Table S5). Moreover, the interaction energy of water molecules with TIP4P/2005 is -11.43 kcal/mol (Table S5), stronger than the experiment of -9.92 kcal/mol.⁴⁹ The ionic HFE is supposed to affect the ion–water interactions largely; there is, however, no consensus to date on this quantity. Therefore, the overestimated ion diffusivity may

result from the imbalanced interactions between ion and water as well.

Nonbonded interaction energy (electrostatics plus vdW) of the ion–water dimer in the gas phase can be used to compare different models.^{29,40,48,81} Figure 6a illustrates the calculation of the interaction energy (IE) as a function of ion–OW distance (IOD), and the IEs between Al^{3+} or Ce^{4+} and one TIP3P water are presented in Figure 6b,c, respectively. IE profiles of the 12-6 HFE set by Li and Merz's group are quite different from that of Li and Merz's 12-6 IOD and 12-6-4 sets and our models. The profiles using the 12-6-4 set and our model are almost overlapped, as indicated by Figure 6 for Al^{3+} and Ce^{4+} , and Figures S10 and S11 (SI) for the other 22 metal cations. Using the 12-6-4 set as a reference, we then computed the minima of IE profiles ($\text{IOD}_{\min}$ for distance and $\text{IE}_{\min}$ for energy) and compared them with the reference. The difference in $\text{IOD}_{\min}$ and $\text{IE}_{\min}$ for our models amounts to approximately zero (Figure 6d,e), indicating an equivalence between our models and the 12-6-4 set by Li and Merz. The 12-6 IOD set generates near-zero $\Delta\text{IOD}_{\min}$ values as well (Figure 6d), while it underestimates the interaction energy between ion and water compared to the 12-6-4 set (Figure 6e). The 12-6 HFE set tends to yield smaller $\text{IOD}_{\min}$ and more negative $\text{IE}_{\min}$ (i.e., stronger interaction) on average (Figure 6).

Osmotic Coefficients of Salt Solution. Our new models targeted two properties of HFE and IOD, and ion–ion interactions were not taken into account in the model development. As a further test, we computed the osmotic coefficients of six trivalent chlorides in aqueous solution with different concentrations (Figure 7a), and the data points are given in Table S6 in the SI. TIP3P-compatible ion models well reproduced the osmotic coefficients of CeCl_3 , LaCl_3 , NdCl_3 , and PrCl_3 solutions, with an RMSE of 0.08 and an MSE of 0.05 on average (Figure 7a and Table 7). However, the osmotic coefficients of AlCl_3 and YCl_3 are underestimated in particular in high concentrations, as revealed by the MSE of -1.40 and -0.15 , respectively (Table 7). A comparison of the cation–anion interacting LJ potentials for different trivalent chlorides indicates that the underestimation appears to result from the small R and large well depth (ϵ) between $\text{Al}^{3+}/\text{Y}^{3+}$ and Cl^- according to the Lorentz–Berthelot (LB) mixing rule (Figure 7b and Table 7).

Although the osmotic pressure of the AlCl_3 solution appears to increase with the increasing salt concentration (Table S6), the LB-based $\text{Al}^{3+}\text{--Cl}^-$ LJ parameters for TIP3P-compatible models lead to a decrease in the osmotic coefficients with the increasing concentration (Figure 7a). Similarly, SPC/E and TIP4P-EW compatible ion models result in small R and large ϵ for cation–anion LJ potentials (Figure 7b) and largely underestimate the AlCl_3 osmotic coefficients. The SPC/E set performs better than the other two sets of TIP3P and TIP4P-Ew (Figure 8a and Tables 7 and S7).

Decreasing the $\text{Al}^{3+}\text{--Cl}^-$ LJ parameters of $R_{\text{Al--Cl}}$ and/or $\epsilon_{\text{Al--Cl}}$ is found to produce a larger osmotic coefficient of 3.0 mol/L AlCl_3 , but it is still an underestimation compared to the experiment (Figure 8b). A further decrease in $R_{\text{Al--Cl}}$ or $\epsilon_{\text{Al--Cl}}$ would lead to unstable simulations, and we instead choose to increase the value of $R_{\text{Al--Cl}}$. An increased $R_{\text{Al--Cl}}$ and a decreased $\epsilon_{\text{Al--Cl}}$ (Figure 8b, bottom right corner) are able to reproduce the experimental osmotic coefficient of 3.0 mol/L AlCl_3 solution. Based on the grid searching of parameter space, we developed two sets of modified $\text{Al}^{3+}\text{--Cl}^-$ LJ parameters (set 1 and set 2, Table 7) and these two sets accurately

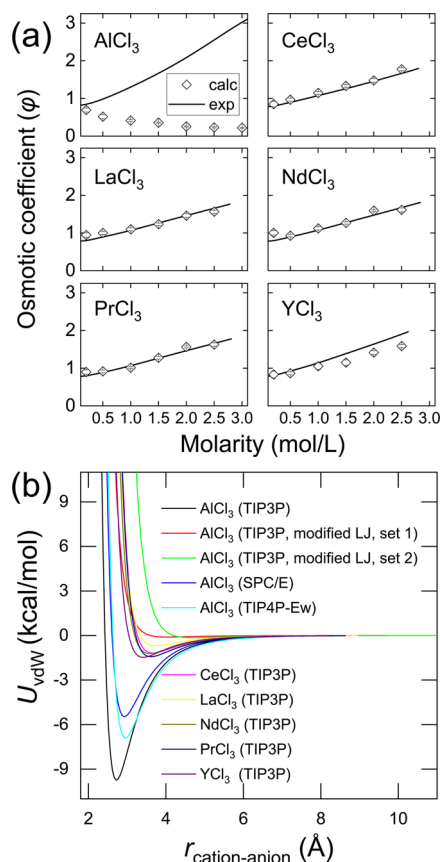


Figure 7. (a) Osmotic coefficients (ϕ) of trivalent chlorides in aqueous solution with different concentrations for TIP3P-compatible ion models. The open diamonds are the calculated ϕ and the solid lines are polynomial fits of experimental observations.^{85,86} (b) vdW potential computed by eq 2 (LB mixing rule) between the trivalent cations and chloride anion as a function of the atom-pair distance (r), except for the modified Al–Cl LJ sets. Water-compatible M^{3+} models (Tables 2–4) and the Joung–Cheatham Cl^- model⁸⁴ were used to describe the trivalent chlorides.

reproduce the osmotic coefficient of the AlCl_3 solution in varying concentrations of 0.2–3.0 mol/L with an RMSE of ~ 0.1 (Figure 8a and Table 7). The calculated osmotic pressure and osmotic coefficients with the modified LJ parameters are given in Table S8, along with the experimental observations.^{85,86} No significant differences are observed in the predicted osmotic coefficients of the AlCl_3 solution by the two methods (cut-off and LJ-PME) for treating vdW interactions (Table S8).

The R parameters of Al^{3+} models amount to ~ 0.2 Å (Tables 2–4), much smaller than its vdW radius of 1.046 Å.⁹² For an improved osmotic pressure, the interacting LJ R parameter between Al^{3+} and Cl^- ($R_{\text{Al--Cl}}$) needs to be tuned to ~ 4.0 Å (Figure 7b and Table 7). The LJ R of the Cl^- model would therefore need to be ~ 3.8 Å, a value much larger than the original Joung–Cheatham model (2.513 Å)⁸⁴ and its vdW radius (2.252 Å).⁹² This implies a transferability issue of using unphysical 12-6 LJ models with a relatively small R and necessitates additional recalibrations.

Metalloprotein Simulation. To assess the performance of our models in biological systems, a metalloprotein of D-xylose isomerase (PDB code: 1XLH) was simulated for 50 ns. This protein has a metal site containing one Al^{3+} ion that coordinates with four amino acid residues and one water

Table 7. Cation–Anion (M^{3+} – Cl^-) LJ Parameters and Comparison of the Calculated Osmotic Coefficients of Salt Solutions with Experiment^a

salt	water	mixing rule ^b	$R_{M-Cl}/2$ (Å)	ϵ_{M-Cl} (kcal/mol)	comparison with experiment ^c		
					RMSE	MAE	MSE
AlCl ₃	TIP3P	LB	1.3625	9.74137	1.66	1.40	−1.40
AlCl ₃	TIP3P	modified LJ, set 1	2.02043	0.105	0.06	0.05	0.04
AlCl ₃	TIP3P	modified LJ, set 2	2.35717	0.15	0.10	0.07	0.07
AlCl ₃	SPC/E	LB	1.46445	5.4645	1.17	0.96	−0.96
AlCl ₃	TIP4P-Ew	LB	1.4764	6.90442	1.53	1.30	−1.30
CeCl ₃	TIP3P	LB	1.83065	1.1797	0.07	0.07	0.07
LaCl ₃	TIP3P	LB	1.87375	0.685994	0.08	0.07	0.03
NdCl ₃	TIP3P	LB	1.7902	1.21834	0.10	0.08	0.06
PrCl ₃	TIP3P	LB	1.80735	1.42748	0.06	0.05	0.03
YCl ₃	TIP3P	LB	1.70725	1.47035	0.19	0.16	−0.15

^aWater-compatible M^{3+} models (Tables 2–4) and Joung–Cheatham Cl^- models⁸⁴ were used to simulate MCl_3 solution at 0.2–3.0 mol/L for $AlCl_3$ and at 0.2–2.5 mol/L for the others. ^bThe Lorentz–Berthelot (LB) mixing rule (eq 2) was used to derive LJ parameters between M^{3+} and Cl^- , except for the two modified LJ sets. ^cRMSE, MAE, and MSE are root-mean-square error, mean absolute error, and mean signed error from experimental osmotic coefficients.⁸⁵

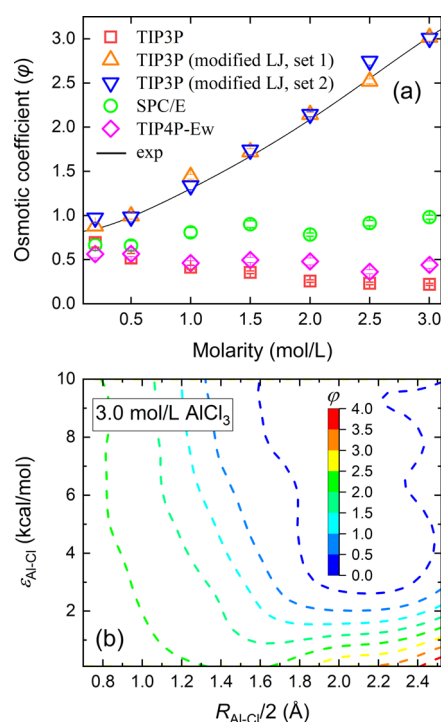


Figure 8. (a) Osmotic coefficients of the $AlCl_3$ solution in different salt concentrations for three water-compatible ion models (TIP3P, SPC/E, and TIP4P-Ew) using LJ parameters between Al^{3+} and Cl^- derived from the Lorentz–Berthelot mixing rule. The results using modified LJ parameters (Table 7) were presented as well. The solid black line indicates experimental observations. (b) Contour plots of osmotic coefficients of 3.0 mol/L $AlCl_3$ solution in the parameter space of $R_{Al-Cl}/2 \in [0.67, 2.53]$ Å and $\epsilon_{Al-Cl} \in [0.1, 10]$ kcal/mol.

molecule in the crystal structure, as shown in Figure 9. Four force field/water combinations of Amber ff14SB/TIP3P, CHARMM36m/TIP3P, CHARMM36m/TIP3P, and a99SB-disp/a99SB-disp were examined using the corresponding water-compatible Al^{3+} models. We did not design metal models for the modified TIP3P water model (i.e., TIP3P) and just adopted the TIP3P-compatible parameters (Table 2) for comparison.

The root mean square deviations (RMSD) of the protein backbone appear stable during the MD simulations and

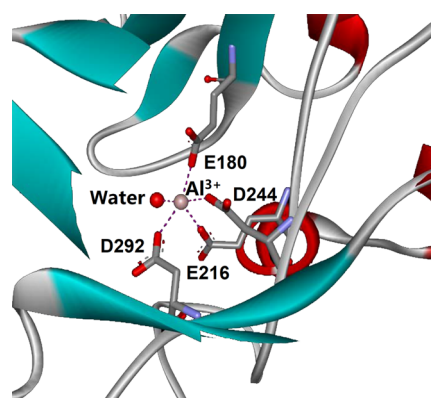


Figure 9. Metal-binding center of D-xylose isomerase in the crystal structure (PDB code: 1XLH). Al^{3+} coordinates with four protein residues (Glu180, Glu216, Asp244, and Asp292) and one water molecule.

fluctuate around 3.1, 3.8, 4.5, and 2.9 Å for the cases of Amber ff14SB/TIP3P, CHARMM36m/TIP3P, CHARMM36m/TIP3P, and a99SB-disp/a99SB-disp, respectively (Figure 10). The RMSD values of the metal-binding residues amount to approximately 0.5, 0.6, 0.6, and 1.3 Å. All of the four FF/water combinations are able to maintain the overall structure of protein 1XLH, and the Amber ff14SB/TIP3P case performs best. There is no obvious difference between TIP3P and TIP3P for use with TIP3P-compatible Al^{3+} models in combination with the CHARMM36m force field. The a99SB-disp/a99SB-disp combination yields the lowest RMSD of the protein backbone, while it gives the largest fluctuation in the metal-binding site.

Interatomic distances of the Al^{3+} -binding coordination spheres were analyzed to pinpoint the details of metal-binding sites (Table 8). In the crystal structure, Al^{3+} binds to two Glu residues (E180 and E216), two Asp residues (D244 and D292), and one water molecule (Figure 9), giving a coordination number (CN) of 5, and the coordinated Glu and Asp residues are in a monodentate binding model. That is, only one of the two carboxy oxygens (OE1 and OE2 for Glu and OD1 and OD2 for Asp) binds to the Al^{3+} ion. The Amber14SB and CHARMM36m force fields reproduce the monodentate binding model accurately, and during the MD

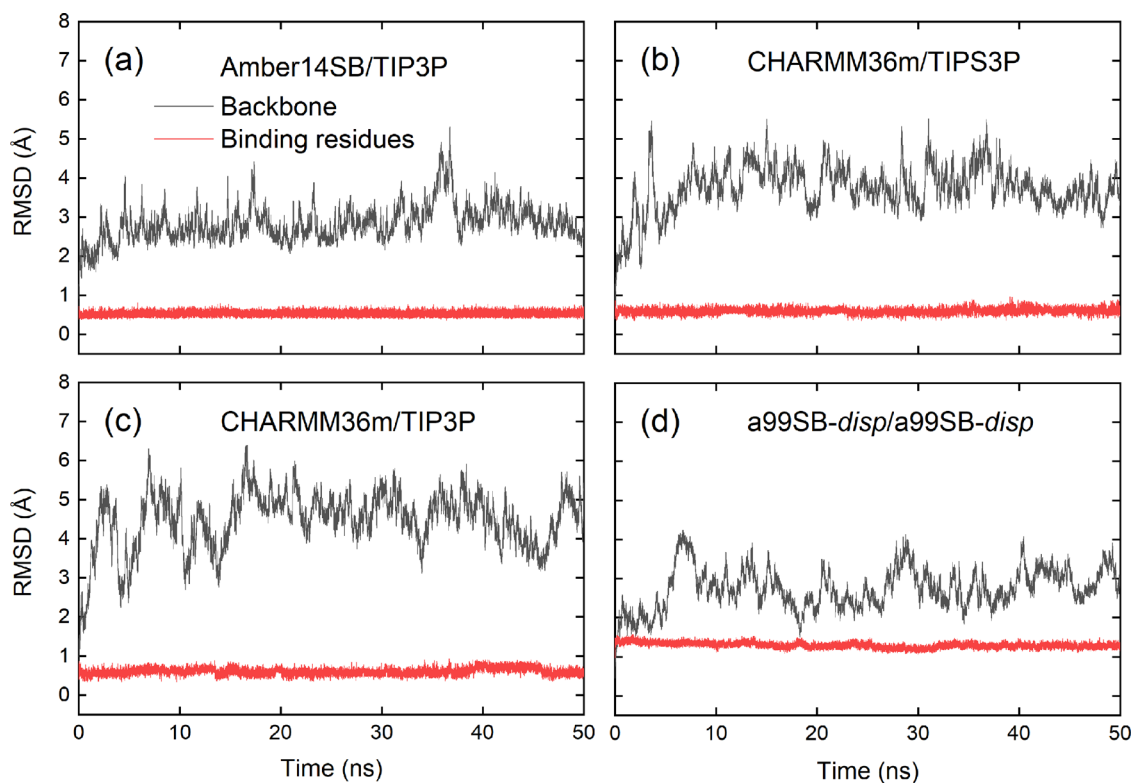


Figure 10. Root mean square deviation (RMSD) of the protein backbone (in black) and metal-binding residues (in red) from the crystal structure of D-xylose isomerase (PDB code: 1XLH) as a function of simulation time. The RMSDs were computed excluding hydrogen atoms.

Table 8. Interatomic Distance (Å) between Al^{3+} and Binding Residues from the MD Simulations for Al^{3+} -Binding Center of D-Xylose Isomerase (PDB ID: 1XLH)^a

force field	water model	E ¹⁸⁰ OE1	E ¹⁸⁰ OE2	E ²¹⁶ OE1	E ²¹⁶ OE2	D ²⁴⁴ OD1	D ²⁴⁴ OD2	D ²⁹² OD1	D ²⁹² OD2	W1	W2
Amber14SB	TIP3P	3.34	1.77	1.76	3.47	3.86	1.77	3.76	1.77	1.91	1.91
CHARMM36m	TIPS3P	3.29	1.78	1.78	3.64	3.83	1.78	3.86	1.78	1.90	1.90
CHARMM36m	TIP3P	3.30	1.78	1.78	3.66	3.85	1.78	3.84	1.78	1.90	1.90
a99SB-disp	a99SB-disp	1.73	1.73	1.72	3.55	3.47	1.72	3.87	1.74	1.96	none
crystal		3.43	2.27	2.12	4.28	4.08	2.01	4.25	2.15	1.97	not exist

^aThe atoms with an Al^{3+} –O distance of >2.5 Å are not considered as a coordinated ligand and are marked in bold; standard deviations for the coordinated atoms amount to 0.02–0.4 Å and that for the noncoordinated are 0.06–0.13 Å; “none” means that the corresponding atom does not coordinate with Al^{3+} during MD simulations; “not exist” means that in the crystal structure only one water molecule coordinates with Al^{3+} and there does not exist one more water completing coordination with a CN of 6.

simulations, one additional water molecule enters into the metal-binding center, completing the coordination spheres with a CN of 6 (Table 8). The a99SB-disp force field predicts a monodentate binding model for E216, D244, and D292 residues, while both the carboxy oxygens of E180 binds to the metal (i.e., a bidentate mode). Such binding does not allow one more water entering into the metal center, although the CN still amounts to 6 (Table 8). Taking the polarization effect into account by for instance a simple modification of atomic charges of Glu carboxyl oxygen atoms might fix this problem.^{34,48,73} The simulated interatomic distances between Al^{3+} and its ligand using our designed Al^{3+} models are smaller than that in the crystal structure by 0.2–0.5 Å. LJ R parameters of the carboxyl oxygen are 1.6612 Å for the Amber14SB and a99SB-disp FFs and 1.7 Å for the CHARMM36m, respectively. These values are smaller than that of the oxygen atom for the used water models: 1.7683 Å for TIP3P and TIPS3P and 1.7763 Å for a99SB-disp. The difference between carboxyl oxygen and water is only about 0.1 Å, and the large

underestimation of cation–amino acid distances probably relates to the transferability issue due to the small R in Al^{3+} models, as mentioned above for AlCl_3 solution simulations.

The hydration free energy (HFE) of metal ions for the ion models by Li and Merz’s group^{30,31} was computed from PME/PBC simulations and matched experimental references of Marcus’s set based on the proton HFE of $\Delta G^\circ(\text{H}^+) = -252.3$ kcal/mol.⁵⁹ Our HFE added a surface contribution to the PME/PBC calculation, leading to a real prediction and, therefore, matched the experiments based on a real $\Delta G^\circ(\text{H}^+)$ of -259.8 kcal/mol.⁶² We chose to compute real HFEs because this was reported to eliminate the difference in the HFE calculation using different methods such as particle- or molecule-based summations and droplet simulations.^{35,62,74,80,99} Both matches appear reasonable and acceptable, as verified by the HFE calculations of charged amino acids and molecular ions.^{79,80} There is indeed a variety of experimental HFEs in the literature, mainly due to the controversial $\Delta G^\circ(\text{H}^+)$ with a difference by dozens of kcal/

mol.^{62,100} Our LJ scanning data were given in the [Supporting Information](#), which would facilitate the model development targeting different HFE references other than the one we used.

CONCLUSIONS

In the present work, we designed nonbonded point charge models of 18 trivalent and 6 tetravalent metal ions using 12-6 Lennard-Jones (LJ) potential in conjunction with 11 water models, namely, TIP3P, OPC3, SPC/E, SPC/E_b, TIP3P-FB, a99SB-disp, TIP4P-Ew, OPC, TIP4P/2005, TIP4P-D, and TIP4P-FB. The designed models simultaneously reproduced hydration free energy (HFE) and ion–oxygen distance (IOD) in the first hydration shell accurately for all of the metal cations with an error within 1 kcal/mol and 0.01 Å on average, respectively. The simulation of Al³⁺-binding center in the D-xylose isomerase from *Arthrobacter* sp. using four force field/water combinations demonstrated the effectiveness of the new models. The accuracy of our models was very similar to the more complex 12-6-4 LJ models parameterized by Li and Merz's group.^{30,31} The C₄ item in the 12-6-4 LJ-type was introduced to consider the overlook of ion-induced dipole interactions, and here we showed that increasing the 12-6 LJ well depth (ϵ) reached the same ends. Compared to Shannon's revised ion effective radii, the LJ R parameters in our models showed an underestimation by ca. 0.2 Å, while Li and Merz's models gave an overestimation by 0.4–0.7 Å. The relatively small R in our models, such as Al³⁺ may cause a transferability issue when interacting with anion ions or amino acids.

The methods of handling vdW interactions using a cut-off with a dispersion correction of energy and pressure or using LJ-PME yielded almost identical predictions on HFE, IOD, ion diffusion, and osmotic coefficient of the AlCl₃ solution. Our new models targeted two properties of HFE and IOD, and the performance of modeling nontargeted ion-pair interactions depends on the kinds of salt solutions. The use of reduced ionic charges plus 12-6 LJ potential was reported to be feasible for point charge models of mono-, divalent-, and highly charged metal ions as well.⁴¹ Given the success of the double exponential function for vdW interactions in the model of divalent metal ions,⁴⁰ such complex functions would be promising in the highly charged ion models. Of these potential functions, the 12-6 LJ form is the simplest one, and its generality and usability need further assessment.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c00723>.

Simulated HFE, IOD, and CN values for optimized three-site (Table S1) and four-site (Table S2) water models; comparison of Lennard-Jones (LJ) R parameters with Shannon's revised ion radii (Table S3); LJ parameters of water oxygen (Table S4), water properties computed using different vdW methods (Table S5); calculated osmotic pressure and osmotic coefficient of salt solution (Tables S6–S8), the optimized metal models (Tables S9–S19) in the form of C_6 – C_{12} , σ – ϵ ; the relationship of HFE (ΔG_{Coul} plus ΔG_{vdW}) and IOD with R and ϵ for the test M(IV) metal ion in the TIP3P water model (Figures S1 and S2), comparison of calculated HFE and IOD with experiment using different vdW methods (Figure S3); relative errors of the

calculated HFE and IOD from experiment for SPC/E (Figure S4); TIP4P-Ew (Figure S5), OPC3 (Figure S6), OPC (Figure S7), TIP3P-FB (Figure S8), and TIP4P-FB (Figure S9) water models, and interaction energy between ion and TIP3P water (Figures S10 and S11) (PDF)

Scanned HFE and IOD with different R – ϵ combinations for the 11 water models (ZIP)

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Notes

The authors declare no competing financial interest.

For the convenience of using our new models, the LJ parameters of M(III) and M(IV) cations are given in the [Supporting Information](#) in the forms of C_6 – C_{12} , σ – ϵ , and R – ϵ . The crystal structure of the D-xylose isomerase from *Arthrobacter* sp. was taken from the Protein Data Bank (<http://www.rcsb.org>). The CHARMM36m and a99SB-disp force fields can be downloaded from http://mackerell.umaryland.edu/charmm_ff.shtml and <https://github.com/paulrobustelli/Force/Fields>, respectively. All of the simulations were implemented using GROMACS 2018 software, except for the water viscosity calculations using GROMACS 5.1.4 (<http://www.gromacs.org>). The AmberTools17 package (<http://ambermd.org>) was used for the preparation of protein simulations with the Amber ff14SB force field.

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